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Received: 28 March 2026

Accepted: 20 June 2026

Published online: 14 July 2026

Cite this article as: Arumalla N. & Gampala V. A Novel Feature Extraction Technique and Hybrid Jaya Skill Optimization with DNF_VGG16 for Brain Tumor Classification Using MRI Image. *Int J Comput Intell Syst* (2026). <https://doi.org/10.1007/s44196-026-01455-0>

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A Novel Feature Extraction Technique and Hybrid Jaya Skill Optimization with DNF_VGG16 for Brain Tumor Classification using MRI Image

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Abstract

A brain tumor (BT) is an unusual and uncontrolled development of cells in the brain. If it is not detected and treated at an early stage, it can progress to a serious condition that may result in death. The current BT classification approaches struggle to handle noisy images and inadequate feature extraction methodologies. Some methods often face issues in handling complex tumor structures, overlapping tissue regions, and intensity levels. To address these limitations, a new Hybrid Jaya Skill Optimization Algorithm (HJSOA) is proposed for classifying BTs from MRI images. This method integrates the Jaya Algorithm with the Skill Optimization Algorithm (SOA). In the preprocessing stage, a Bilateral filter is applied to medical images to remove noise and artifacts, thereby improving image quality and enhancing classification accuracy. Weighted Fuzzy K-means (WFKM) clustering is used to segment the pre-processed images. This approach enables accurate delineation of BT boundaries, thereby enhancing overall performance and minimizing false detections. In addition, Discrete Wavelet Transform (DWT)

with the Local Texton XOR Patterns (LTxXORP) is utilized to derive the features. It helps to handle complex and subtle patterns in the input data, which leads to faster computation and improves the generalization of the model. Finally, the BT is classified using the Deep Neuro Fuzzy–Visual Geometry Group 16 (DNF_VGG16) model, which is formed by combining the Deep Neuro-Fuzzy Network (DNFN) with the Visual Geometric Group (VGG16) architecture. Finally, the classification of BT is implemented using the Hybrid Jaya Skill Optimization Algorithm (HJSOA) and **DNF_VGG16**. The effectiveness of the proposed HJSOA+DNF_VGG16 model is evaluated by using different training configurations on the BRATS 2018 and Figshare datasets. In the BRATS 2018 dataset, the developed HJSOA+**DNF_VGG16** model attained maximum values of **0.930 for accuracy**, **0.950 for Total Positive Rate (TPR)**, and **0.940 for Total Negative Rate (TNR)** at the learning sets of 90. The performance enhancement achieved by the developed model based on accuracy is 17.0%, 14.45%, 11.6%, 7.84%, 4.90%, **3.50%**, **1.58%**, **0.75%**, and **0.39%** higher than the Deep Convolutional Neural Network (DCNN), Multi-task Attention Guided encoder-decoder Network (MAG-Net), Hybrid Convolutional Neural Network and support vector Machine (CNN-SVM), Deep Neural Network (DNN), conditional Aquila horse herd optimization driven deep neuro-fuzzy network (CAHO-based DNFN), **ResNet101+Grey Wolf Optimizer (GWO)+Multi-Transformer Fusion model**, Deep Neuro Fuzzy-Visual Geometric Group-16 (DNF_VGG16) models, **ConvNeXt Base model**, and **Transformer□Guided Cross□Scale Attention and Deep Semantic Fusion network (TCAFNet)**.

Keywords: MRI image; Weighted Fuzzy K-Means Clustering; Hybrid Jaya Skill optimization algorithm; brain tumor classification; Deep Neuro Fuzzy-Visual Geometry Group-16.

1. Introduction

The brain includes nearly 100 billion nerve cells, which is considered the vast and intricate organ that manages the entire nervous system as well as is **also** referred to as the crucial organ, which is initiated at the middle part of the nervous system. So, several forms of irregularity that are present in the human brain may severely damage the entire brain system. Despite its complexity, the brain is vulnerable to various disorders, including brain tumors (BTs), which can severely disrupt its function. Primary tumors originate in the brain tissue, while secondary tumors spread to the brain from other parts of the body via the bloodstream [1]. Meningioma, as well as Glioma, are the two fatal kinds of BTs between the primary tumors, which may cause a person to die if it is not established in the first phase. Glioma is one of the most common types of BTs in humans [2] [3].

BTs are classified into four grades. Here, lower-grade tumors such as meningiomas are described in Grade 1 plus Grade 2, whereas Glioma is included in Grades 3 and 4. In medical practice, the incidence rates of meningioma, glioma, as well as pituitary are about 15%, 45%, and 15%, **respectively** [4]. MRI is a sophisticated method that is employed of **medical** imaging that is employed to generate an image of a human body part in high quality, and also it is used frequently while treating BTs, feet, and ankles [5]. The complete anatomical information can be derived from these high-resolution images to observe the abnormalities as well as the development of the human brain [6]. Mainly, the detection of early BT depends on the skill of the radiologist [7] [8]. The tumor diagnostics will be done after confirming the output whether is normal or not [9]. Normally, a biopsy is performed to examine whether the tissue is benign or malignant [10]. It is vital to develop an efficient **diagnostic** tool for both tumor segmentation and classification from MRI images to acquire exact diagnostics and to evade surgery as well as subjectivity [11]. A sequence of physical and neurological assessments is executed to analyze the BTs. Diagnosis is done with Magnetic Resonance (MR), Computerized

Tomography (CT), as well as pathological estimation is performed to validate the diagnosis [12] [13]. MR is the most preferred one among all the imaging modalities since it is considered the only non-invasive as well as the non-ionizing modality [14]. As a consequence, phase, as well as the category of a tumor, is accurately as well as preparing the plan for further treatment. In this process, the manual analysis of medical images consumes lots of time as well as it produces faulty consequences [15]. So, these kinds of problems are solved by an automated Computer-Aided Diagnostic (CAD) system to reduce the workload of diagnosis and classification of brain MRI, such that it performs as a tool to aid radiologists [4].

The powerful tools are yielded through the development of computers that assist in attaining more precise diagnoses [16]. The techniques of BT detection are based on two types: machine learning (ML) and deep learning (DL) [17] methods [18]. Mostly, ML-based techniques and conditional random forest (RF) require hand-crafted characteristics. In this, the hand-crafted factors are essential to be removed from the training images to begin the procedure of learning as well as it requires specialists with wide information to detect the most significant factors [19]. The accuracy is partial while dealing with huge databases. In the meantime, DL-based algorithms have revealed high performance in different industries including medical imaging [20]. The medical images are classified in a better way with the rapid growth of DL approaches and their ability [21]. Among the specialists in medical imaging, Computer-aided diagnosis (CAD) has become a more extensively applied technique of diagnosis [16]. The most familiar DL model is the Convolutional Neural Network (CNN) [22] [23], which automatically finds out the intense features directly from the training data owing to its nature of weight-sharing [20][24]. It is essential to pre-process the MR images before integrating them with various networks to attain the required outcome of classification. The CNN is utilized to solve this difficulty through the desires of feature engineering [25]. The

limitation of resources delays the system utilization for medical diagnostics, and the system must be related to brain disorders for regular clinical diagnosis [26].

1.1 Research Gaps and Motivation

The classification of BT is vital for the precise treatment for saving the lifespan of the patient since it is considered a fatal disease. Classifying BTs manually using MRI scans takes considerable time and often leads to mistakes due to human involvement. The current BT classification approaches struggle to handle noisy images and inadequate feature extraction methodologies. The existing approaches often suffer from low accuracy and poor generalization across different datasets due to limited feature extraction and overfitting. Traditional methods often face issues in handling complex tumor structures, overlapping tissue regions, and intensity levels. Most of the methods lack interpretability and robustness, which makes them difficult to validate and apply in clinical settings. Since numerous techniques are trained specifically on certain datasets, they tend to perform poorly on unseen samples. To mitigate this limitation, an HJSOA+DNF_VGG16 framework is introduced for BT classification. The developed model offers a highly automated, efficient, and accurate classification approach that enhances the diagnostic results of the model.

1.2 Contribution

The present research developed the HJSOA for the classification of BT. The process begins with applying a bilateral filter to the input image for pre-processing, followed by image segmentation using the WFKM clustering technique. These segmented images are then augmented, and the features, namely DWT with LT-XORP in the feature extraction. Then, it is classified with the DNF_VGG16.

The main contribution of the present research is,

- To achieve effective BT classification from MRI images, a hybrid DL framework named HJSOA+DNF_VGG16 is introduced by merging DNFN with VGG16, thereby strengthening feature learning and predictive performance.
- By integrating the powerful feature learning mechanism of VGG16 with the adaptive reasoning capabilities of DNFN, the proposed DNF_VGG16 framework achieves superior classification accuracy and distinguishes tumor characteristics more effectively than traditional CNN models.
- A novel hybrid optimization algorithm, HJSOA, combining the Jaya Algorithm and SOA, is introduced to optimize the training process of the DNF_VGG16 model, which improves convergence behavior and overall classification performance.
- The proposed framework enhances overall tumor classification accuracy and robustness and demonstrates improved performance over existing hybrid CNN-based BT classification approaches.

1.3 Organization

The subsequent section of the entire research is structured such that the analysis and limitations of the BT classification are examined in [Section 2](#), and detailed information regarding the developed HJSOA technique is mentioned in [Section 3](#). Lastly, Sections 4 and 5 depict the analysis results and [the](#) conclusion of the entire research.

2. Literature Survey

Irmak, E., [27] presented the Convolutional Neural Network (CNN), which was assisted by radiologists and physicians for their validation of initial screening. This method improves precision and allows customization for different diagnostic needs. The system was designed to aid medical experts and radiologists by providing validation for their initial BT screening

outcomes. However, the great hardware and the hyper-parameters of the well-known system **had** to be optimized to minimize the computational cost. Rasheed, Z., *et al.*, [26] developed the Gaussian-blur-based sharpening and Contrast-Limited Adaptive Histogram Equalization (CLAHE) with CNN, which **resides** in its ability to significantly reduce the necessity for manual actions, like feature extraction as well as tumor localization. The method was designed to perform reliable classification of brain MRI scans into categories such as glioma, pituitary tumor, and non-tumor (normal) cases. It automated the BT detection process and minimizes the time required for manual MRI analysis. However, this was not only time-intensive but also vulnerable to imprecision and intended to revolutionize the methods of medical imaging.

Alhassan, A.M. and Zainon, W.M.N.W., [28] introduced the Hard Swish-Based RELU Activation Function-CNN Model. This technique follows a three-phase workflow that includes pre-processing the images, extracting relevant features, and performing classification. It reduced the problem of overfitting and reached the best performance when no manual segmentation was required. This approach was capable of handling large volumes of MRI images in a short amount of time thereby enhancing the system's performance. However, the clustering-based segmentation algorithm should be included to enhance BT classification.

Gab Allah, A.M., *et al.*, [16] designed the VGG19 + CNN and Progressive Growing Generative Adversarial Network (PGGAN), which exhibited better generalization and was capable of generating better results with a limited number of images. This approach performed BT classification on MRI images by employing VGG19 for feature extraction and integrating it with three different classifiers. It helped in the early and precise identification of BT types and attained high classification accuracy. Still, it did not acquire the main images with suitable dimensions to create the images as sensible as possible.

Kang, J., *et al.*, [2] introduced the technique of a novel feature ensemble, which produced better performance, especially for larger datasets. This method supported early diagnosis of tumor in

all age groups which helps to provide timely and effective treatment plans. However, the model size must be reduced to enable real-time medical diagnosis through the use of information distillation techniques. Ozkaraca, O., *et al.*, [4] designed the architecture of a dense CNN, which enhanced the feature propagation, mitigated the fading gradient problem, promoted feature reuse, and minimized the number of parameters. This method reduced the overfitting issue and generalizes well across new or unseen data. Here, the processing time was higher and it provided a late response since the model was formed with both dense as well as convolutional networks. Masood, M., *et al.*, [20] developed the Mask Region-based Convolution neural network (Mask RCNN) that evaluated the deep features with efficient representations of BTs. This approach achieved high performance despite having limited data. However, the robustness of the developed scheme was not computed for the analyses of other medicinal image applications like COVID-19 detection, skin cancer, as well as finger skin recognition. Alanazi, M.F., *et al.*, [10] introduced the developed transfer deep-learning model for the diagnosis of BTs in an earlier stage. This mode differentiated tumor subclasses, which helps to precise and early diagnosis of the diseases. This scheme reduced the training time as well as resolved the issues of adaptability, but it was not executed in real-time as well as generalization capability was less. Aamir., *et al.*, [29] suggest a CAD system based on DL to detect brain cancer using MRI images. This method can effectively identify the BT while reducing the overfitting issue. However, the implementation of this method was computationally expensive. Aamir., *et al.*, [5] generated an automated technique called EfficientNet-B0 and ResNet50 for the detection of BT using MRI images. It can detect the tumor even with varied tumor sizes, shapes, and locations. This approach minimized the dependence on human experts for the identification of BT. This approach does not differentiate the type of BTs. Hassan, N.M.H. and Boulila, W., [30] proposed a novel framework that integrated fuzzy logic-based segmentation with DL techniques to enhance BT detection and classification in MRI scans. The method was

developed to enhance diagnostic accuracy by integrating fuzzy logic with a CNN model for improved tumor detection and classification. This model accurately estimated tumor size, which provided valuable clinical insights. However, the framework was not implemented in hardware systems to further optimize its real-time capabilities and integration into medical imaging workflows. Table 1 provides an overview of the analysis conducted on the literature survey of the proposed model.

Table 1. Literature Summary

Author Name	Methods	Advantages	Disadvantages
Irmak, E., [27]	CNN	This method attained high precision and allowed the model to adapt to different diagnostic needs.	It requires great hardware and hyperparameters to optimize the computational cost.
Rasheed, Z., <i>et al.</i> , [26]	Gaussian-blur-based sharpening and CLAHE with CNN	This method reduces the manual action and automates the BT detection process.	It is time-consuming and vulnerable to imprecise and intended to revolutionize the methods of medical imaging.
Alhassan, A.M. and Zainon, W.M.N.W., [28]	Hard Swish-Based RELU Activation Function-CNN Model	It reduced the overfitting issue and reached the best performance when no manual segmentation was required.	It required cluster-based segmentation for a more accurate classification of BT.
Gab Allah, A.M., <i>et al.</i> , [16]	VGG19 + CNN and PGGAN	It exhibits better generalization and can generate better results with a limited number of images.	This model did not capture the main images with suitable dimensions to create the images as sensible as possible.
Kang, J., <i>et al.</i> , [2]	A novel feature ensemble technique	It attained better performance, especially for larger datasets.	To support real-time medical diagnosis, the model must be reduced in size through the application of information distillation methods.
Ozkaraca, O., <i>et al.</i> , [4]	Dense CNN	This method reduces the overfitting issue and generalizes well across new or unseen data.	The processing time was higher and it provided a late response.

Masood, M., <i>et al.</i> , [20]	Mask RCNN	This method attained high performance even with limited data.	This scheme was not computed for the analyses of other medicinal image applications like COVID-19 detection, skin cancer, as well as finger skin recognition.
Alanazi, M.F., <i>et al.</i> , [10]	Transfer deep-learning model	It can also differentiate between tumor subclasses, which helps to precise and early diagnosis of the diseases.	It was not executed in real-time as well as generalization capability was limited .
Aamir., <i>et al.</i> , [29]	CAD system based on DL	This method can effectively detect the BT and minimizes the overfitting issue.	Implementation of this method is computationally expensive, especially in resource resource-constrained environments.
Aamir., <i>et al.</i> , [5]	EfficientNet-B0 and ResNet50	It can detect the tumor even with varied tumor sizes, shapes, and locations	This method does not distinguish the type of BTs.
Hassan, N.M.H. and Boulila, W., [30]	fuzzy logic with CNN	This model accurately estimated tumor size, which provided valuable clinical insights.	This method was not implemented in hardware systems to further optimize its real-time capabilities.
Proposed	HJSOA+DNF_VG G16	The approach allows for effective identification of BTs in the early stages.	It lacks strong generalization capability across different medical image types, including CT and endoscopic scans.

3. Proposed HJSOA for BT classification

This research develops a new model of HJSOA for classifying BT. Initially, the input of the MRI image is taken from two databases, such as [31] [32] and then it passes to the stage of image preprocessing, in which the Bilateral filter [33] is used to eliminate the noises that are present in the image. Then, the pre-processed image is directed to the segmentation phase, where the region of the BT is segmented with the Weighted Fuzzy K-means (WFKM) [34] scheme. Following segmentation, the data augmentation is performed using techniques, such as Rotation, colour augmentation, scaling, contrast, and image translation. Subsequently, the feature extraction process is performed, which is done based on Discrete Wavelet Transform

(DWT) [35] with the Local Texton XOR Patterns (LTxXORP) [36], in which the features are extracted. Finally, the classification of BT is achieved with the Deep Neuro Fuzzy-Visual Geometric Group-16 (DNF_VGG16), which is trained with the developed HJSOA. Here, the DNF_VGG16 is generated by integrating the Deep Neuro-Fuzzy Network (DNFN) [37] and Visual Geometric Group-16 (VGG16) [38], wherein the proposed HJSOA is developed by the combination of the Jaya Algorithm [39] and SOA [40]. Thus, Figure 1 showcases the structural diagram of the developed HJSOA model.

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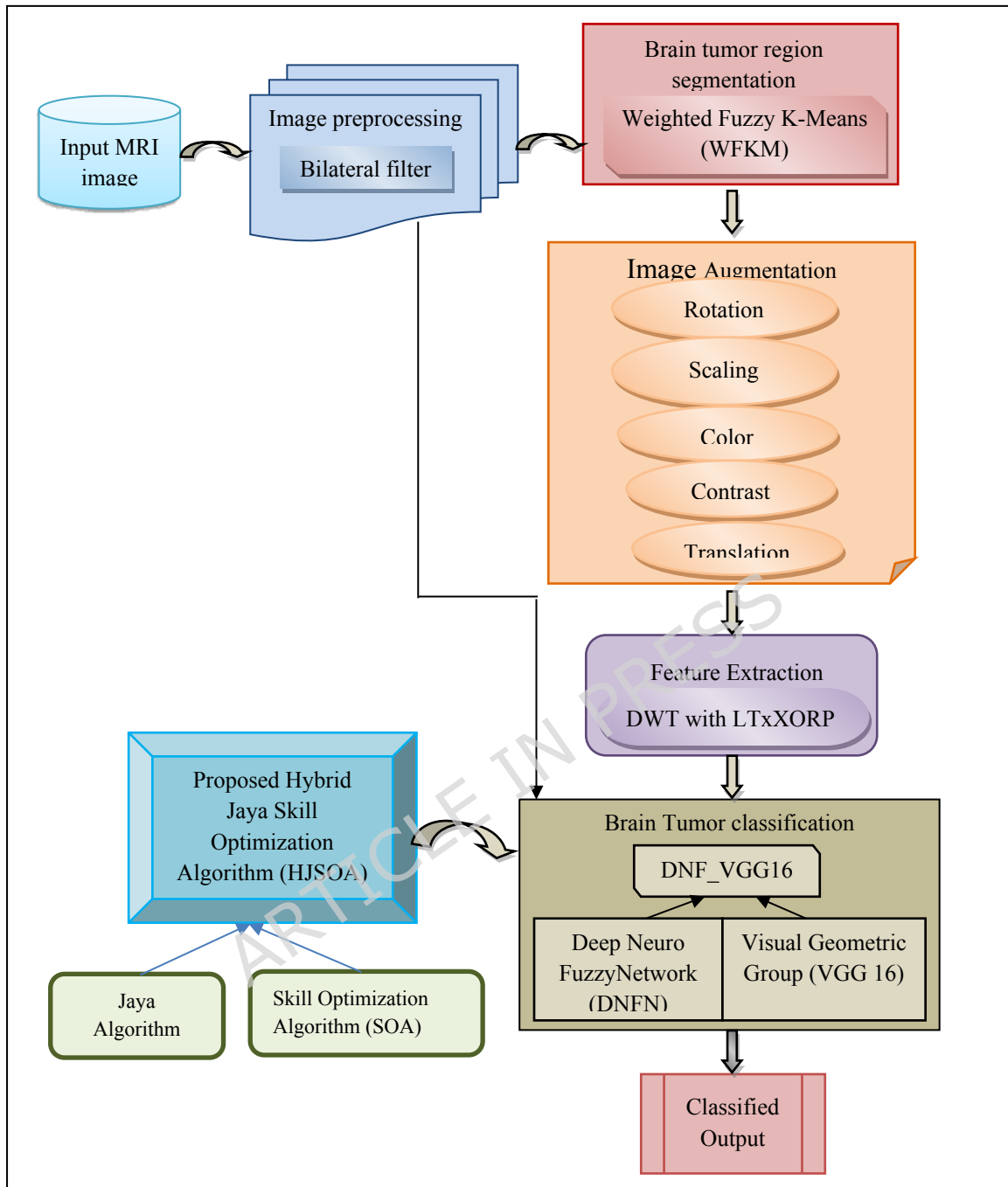


Figure 1. Block diagram of the HJSOA for BT classification

3.1 Image Acquisition

The dataset considered for BT classification is denoted as D , which is used for acquiring images and is depicted as,

$$D = [I_1, I_2, \dots, I_m, \dots, I_n] \quad (1)$$

where, D shows the input image dataset. m^{th} image is represented as I_m , which signifies the input image of BT classification, and n^{th} image is denoted as I_n , which denotes the entire image of a dataset D .

3.2 Image Pre-processing

The input image I_m is passed to the phase of image pre-processing, which eliminates the unwanted noise through the Bilateral filter [33] that is described in the [section below](#).

3.2.1 Bilateral filter

Usually, the Bilateral filter (BF) [41] can smooth the images while conserving edges through the nonlinear combination of nearest values as well as being approximately like a Gaussian filter, which is defined as,

$$BF(I_m)_p = \frac{1}{W_p} \sum_{i \in S} U(x, i) I_i \quad (2)$$

where, $BF(.)$ represents the Bilateral filter, I_m and I_x specifies the input image and value at the position of the pixel x . S signifies the spatial domain and $\sum_{i \in S}$ specifies the sum over [all](#) pixels of the image indexed by i . The 2-D Gaussian kernel is represented as $U(x)$, which follows as,

$$U(x) = \frac{1}{2\pi\sigma^2} \exp\left(-\frac{x^2}{2\sigma^2}\right) \quad (3)$$

where, the neighborhood size is defined by the parameter σ . The normalization constant W_p is expressed as,

$$W_p = \prod_{i \in S} U_{\sigma_s}(\|x - i\|) U_{\sigma_r}(\|x - i\|) I_i \quad (4)$$

where, σ_s and σ_r represent the spatial as well as range factors for spatial Gaussian weighting U_{σ_s} and range Gaussian weighting U_{σ_r} . $\|x - i\|$ denotes the Euclidean distance among the location of the pixel x and i . Thus, the BF is considered as the combination of both spatial as well as range filters. The output from this stage is denoted as I_p .

3.3 BT Region Segmentation

The pre-processed image I_p is fed into the phase of segmentation. Usually, the image segmentation splits the digital image into numerous regions depending on the features of an image pixel. Here, the region of the BT is segmented through the WFKM [34] scheme.

3.3.1 Weighted Fuzzy K-Means technique for BT segmentation

The foremost intention of the WFKM scheme is to fuzzify the values of pixels with the assigned weights for identifying the tumorous area in a precise manner. The WFKM approach merges fuzzy c-means with k-means while using spatial framework information to guide clustering. The difference in illumination as well as the grey values of a pixel is controlled through the scheme of penalized membership while clustering. Due to FCM, this scheme manages the scattered clusters as well as unclear segments during iteration. WFKM functions to the minimization of the fitness function $F(C, U)$ that is expressed as,

$$F(C, U) = \sum_{a=1}^n \sum_{b=1}^c u_{ab}^m \|I_a - C_b\|^2 + \frac{1}{2} \sum_{b=1}^c |l_b - w_b|, 1 \leq m \leq M \quad (5)$$

where, u_{ab}^m denotes the membership degree of the pixel I_a in the cluster C_b . λ represents the constant that manages the impact of illumination, l_b and w_b is referred to as the luminosity and weight, which is computed for the cluster b that is expressed as,

$$l_b = \frac{\sum_{a=1}^n n I_a^b}{\sum_{a=1}^n u_{ab}} \quad (6)$$

where I_a^b signifies the image's a^{th} pixel belongs to b^{th} cluster, u_{ab} represents the degree of membership of the pixel I_a within the cluster C_b , and n indicates the total number of pixels.

$$w_b = \frac{\sum_{a=1}^n \sum_{d=1}^n |I_a^b - I_d^b|}{\sum_{a=1}^n u_{ab} \sum_{d=1}^n u_{ab} + 1} \quad (7)$$

where I_d^b represents the image's d^{th} pixel belonging to b^{th} cluster, u_{ab} denotes the degree of membership of the pixel I_a in the cluster C_b .

The pixel's membership I_a in b^{th} cluster is updated as,

$$u_{ab} = \frac{1}{\sum_{d=1}^c \left(\frac{|I_a - C_b|}{|I_a - C_d|} \right)^{\frac{2}{m}}}, \quad \text{if } a \text{ and } b \neq a \neq n; \quad (8)$$

where, C_b and C_d denotes the center of b^{th} and d^{th} cluster. m represents the exponent of fuzziness.

Thus, the output of these segmented images is referred to as I_s , and passed to the next phase.

3.4 Image Augmentation

The segmented output I_s is taken as an input for image augmentation [42], which is employed to increase the diversity of the dataset by generating additional variations of the existing images. Before dividing the dataset into training and testing sets, image augmentation was carried out exclusively on minority class data to address imbalance, thereby increasing data diversity and supporting better model training through varied image inputs. This data augmentation is executed with techniques such as Rotation, Scaling, Color augmentation, Contrast, and Translation, which are described below,

3.4.1 Rotation

The image is positioned at 90° or adjusted to slight angles according to preferences. Following this, no **noise** is present in the relevant image when it is tilted 90° . Thus, the parameter of the tilted degree will establish the security of the augmentations, which is represented as I_r .

3.4.2 Scaling

The image can be enlarged through the scheme of scaling, which is considered a form of usual arithmetical image augmentation that contains two varied schemes. An image is separated from its normal position, and then it is processed back to its normal dimension. Thus, the scaled image is denoted as I_s .

3.4.3 Color augmentation

The color augmentation alters the pixel value without altering its position, and also it modifies the intensity of the channels through the variation present in images. Therefore, the augmented color image is signified as I_a .

3.4.4 Contrast

Contrast refers to the variation between dark and light regions in an image; however, it does not categorize the image based on contrast levels. Also, it specifies the possible strength. Hence, the contrasted image is indicated as I_c .

3.4.5 Translation

Translation is used to locate the object within the image, allowing the system to scan the entire image, including background noise and the output is denoted as I_t .

Therefore, the final output obtained through the augmented data is expressed as,

$$A_g = [I_r, I_s, I_a, I_c, I_t] \quad (9)$$

3.5 Feature extraction

The augmented output A_g is further extracted through this feature extraction process with the DWT-LTxXORP. Initially, when the augmented image is taken and applied to DWT, then it is split into four bands Low Low (LL) band, Low High (LH) band, High Low (HL) band, as well as High High (HH) band. Then LTxXORP is applied only to three bands, such as LL, LH, and HL to attain three texture features such as f_1^1 , f_1^2 , and f_1^3 as well as it is not applied to the HH band since the HH band consists of high noise. Subsequently, the histogram is applied to these texture features f_1^1 , f_1^2 , and f_1^3 to generate the Feature vector F . The entire process of feature extraction is represented in Figure 2

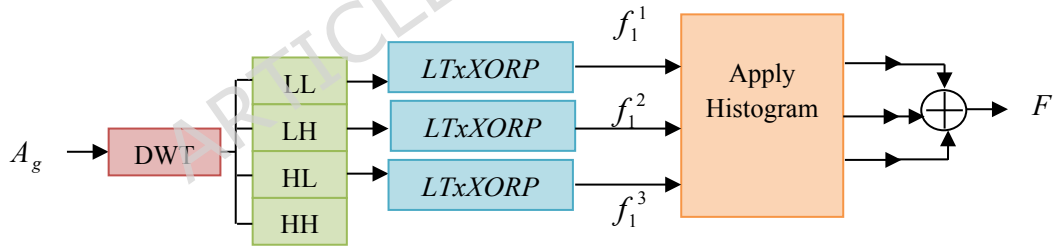


Figure 2. Process of feature extraction

3.5.1 DWT features

The two-dimensional DWT [35] is considered since it decomposes the images on various frequency levels. In addition, the DWT presents multi-resolution as well as location information as a type of wavelet transform. It performs low-pass and high-pass filters on similar images from both vertical and horizontal dimensions by assuming the first-level DWT. Then, the concerned image is split into 4 blocks, such as LL, LH, HL , and HH . The block LL denotes

the coefficients of coarse-scale DWT, while the blocks LH , HL and HH denote the coefficients of fine-scale. Particularly, LL , LH , HL and HH is referred to as an estimated element, horizontal element, vertical element, as well as diagonal element, correspondingly. The LL sub-band required should be filtered until it reaches the last level D to attain a higher level of DWT. Finally, the image includes LL_D and LH_X , as well as HL_X and HH_X , in which the value of X ranges from 1 to D . The output of DWT is passed to $LTxXORP$.

3.5.2 DWT with $LTxXORP$

The output of DWT features is considered as an input $LTxXORP$. Here, the seven various shapes of Texton [36] are considered for the generation of Texton images, which is shown in Figure 3.

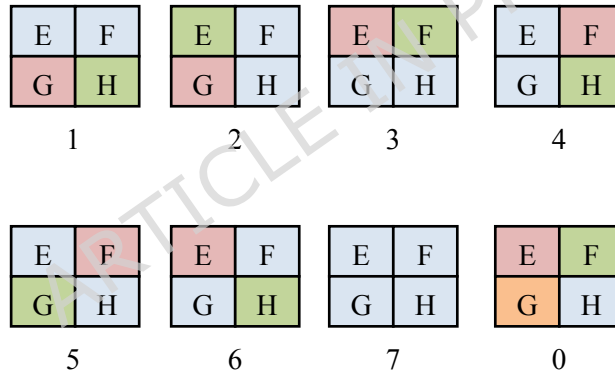


Figure 3. Texton shape for feature extraction

Assume that the image is split into overlapping 2×2 subblocks such as I . The position of gray values is referred to as " E, F, G, H ", which is coded depending on the shape of the texton that is given as,

$$T_x[x, y] = \begin{cases} 1, I(E) \oplus I(F) \& I(G) \oplus I(H) \\ 2, I(F) \oplus I(H) \& I(E) \oplus I(G) \\ 3, I(G) \oplus I(H) \& I(E) \oplus I(F) \\ 4, I(E) \oplus I(G) \& I(F) \oplus I(H) \\ 5, I(E) \oplus I(H) \& I(F) \oplus I(G) \\ 6, I(F) \oplus I(G) \& I(E) \oplus I(H) \\ 7, I(E) \oplus I(F) \oplus I(G) \oplus I(H) \\ 0, I(E) \oplus I(F) \oplus I(G) \oplus I(H) \end{cases} \quad (10)$$

After the evaluation of the texton image, the center of each pixel executes the XOR operation. Thus, the LTxXORP is coded as,

$$LTxXORP_{Q,R} = \sum_{n=1}^Q 2^{n-1} \cdot f_3[T(g_n) \oplus T(g_c)] \quad (11)$$

$$f_3[T(g_n) \oplus T(g_c)] = \begin{cases} 1 & T(g_n) \oplus T(g_c) \\ 0 & \text{else} \end{cases} \quad (12)$$

where, $T(g_n)$ and $T(g_c)$ specifies the form of the texton for the neighbor pixel g_n as well as the center pixel g_c . $\oplus$ denotes the XOR function among the variables. The entire map after the evaluation of LTxXORP is symbolized by constructing a histogram as,

$$H_{LTxXORP}(l) = \sum_{o=1}^{N_1} \sum_{p=1}^{N_2} f_2[LTxXORP[o, p] \setminus l] \cdot 2^p \cdot 1 \quad (13)$$

Finally, the feature vector F is expressed as,

$$F = f_1^1 \oplus f_1^2 \oplus f_1^3 \quad (14)$$

where f_1^1 represents the texture features from LL band, f_1^2 denotes the texture features from LH band, and f_1^3 shows the texture features from HL band,

3.6 BT classification

The outcome of the extracted features F is considered as the input for BT classification. Here, DNF_VGG16 is utilized to improve the performance while classifying BT, which is generated

by integrating two networks, such as DNFN [43] and VGG16 [44]. The DNF_VGG16 architecture for BT classification is shown in Figure 4. Initially, when I_p is assumed as an input and passed to VGG16, then it attains the output H_1 . This acquired output H_1 and the extracted feature F is subjected to the module of DNF_VGG16. Here, it performs the process of fusion depending on the regression modelling and generates the output H_2 . Subsequently, the resulting output H_2 is acquired by the DNFN layer and generates the output H_3 .

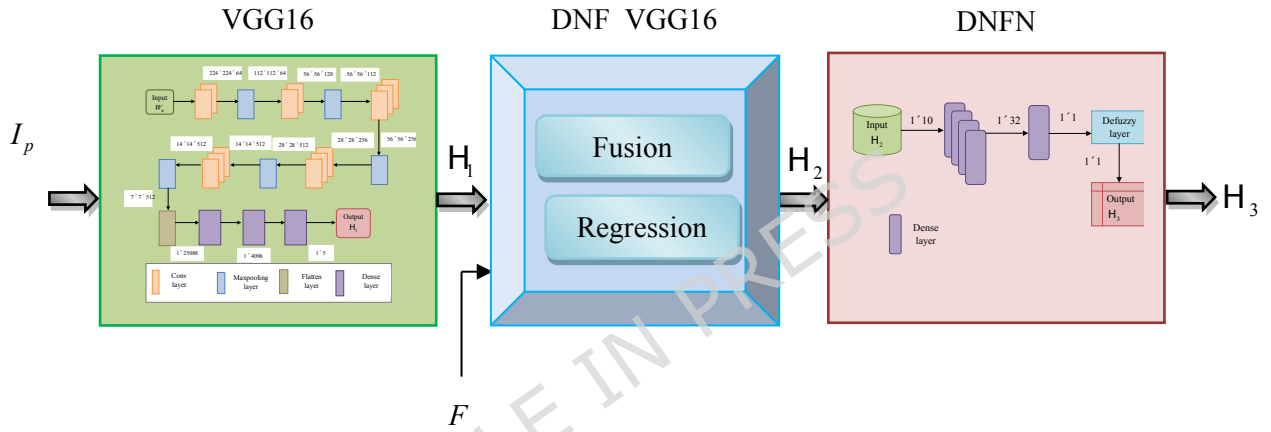


Figure 4. The general architecture of DNF_VGG16

3.6.1 VGG16 model

VGG16 [44] is a form of CNN model that consists of 16 layers and is efficiently utilized for several applications, like object detection, categorization, image segmentation, and so on. Several CNN approaches are initiated to attain the best consequences. Here, I_p is assigned as an input, and several layers of VGG16 are described below.

i) Conv 2D layer

It is used to select relevant features from the input image through multiple interconnected layers that pass information sequentially from one layer to the next. This layer demonstrates the non-linear mapping, which is represented as,

$$n \ll E^r M \quad (15)$$

where, E^r and M represents the input filter matrix and weight, and n depicts the layer's output.

ii) Max pooling

This layer reduces the feature map resolution, and the kernel helps confirm the key features present in each channel. Therefore, it is represented as,

$$P_{\max}(i) \leq P_{\max}(i) \quad (16)$$

where $P_{\max}$ denotes the max pooling, $i[\cdot]$ represents the input vector, and $\cdot$ signifies the input vector.

iii) Flatten layer

It is employed to change the feature map output, with its main function being the conversion of matrix elements into vectors.

iv) Dense Layer

This is considered the FCL that develops the linear process to replace the image from the preceding layer to an ordered shape. Thus, the outcome of VGG16 is,

$$H_1 \square S \square G_{\frac{1}{n}} \square I_{\frac{1}{n}} \square u_n E \square M_2 \square p \square M_2 \square r_n \square M_{u_n} \quad (17)$$

where, H_1 signifies the VGG16, S and G denotes the matrix weight, $\square$ represents the FCL, I_p denotes as input, $\square$ represents the FCL, and $\square$ represents the convolution.

3.6.2 DNF_VGG16 layer

The extracted features F and the attained output such as H_1 are referred to as the input for the layer of DNF_VGG16. Depending on the regression, the output of VGG16 with the mined features is performed using the Fractional Calculus (FC) [45]. FC boost computational efficiency and decrease information loss present in the DNF_VGG16 architecture. FC used in

the proposed fusion mechanism introduces memory-based feature interaction between VGG16 and DNFN outputs. Unlike conventional fusion methods such as concatenation or weighted addition, which consider only current feature values, fractional-order modeling incorporates historical feature dependencies through a non-local operator. This allows the model to capture long-range relationships between extracted features, leading to a more informative and adaptive fusion process. By the consideration of output at several time instances, the fusion is accomplished.

The general equation for FC is expressed as,

$$H(t+1) = A * H(t) + \frac{1}{2} A * H(t-1) \quad (18)$$

The output at time t is obtained based on the extracted features F and is expressed as

$$H(t) = \sum_{i=1}^n F_i * b_{di} \quad \text{and at time } t-1, \text{ the output is based on the VGG16 output } H_1. \text{ Thus, the}$$

attained output H_2 is expressed as,

$$H_2 = A \sum_{i=1}^n F_i * b_{di} + \frac{1}{2} A * H_1 \quad (19)$$

where, A denotes the constant, b_{di} denotes the weight, F_i represents the feature vector.

Now, substitute eqn (17) in (19),

$$H_2 = A \sum_{i=1}^n F_i * b_{di} + \frac{1}{2} A * (S * G_{\text{VGG16}} * \text{ReLU} * \text{ReLU} * \text{ReLU} * E * \text{ReLU} * \text{ReLU}_2 * \text{ReLU}_1 * \text{ReLU}_p * M_2 * \text{ReLU} * M_{u_{\text{VGG16}}}) \quad (20)$$

3.6.3 DNFN model

The output H_2 generated by DNF_VGG16 is processed through the DNFN model, which is associated with the cascaded system. The learning mechanism is incorporated with the DNFN vector, and it can be represented as,

$$H_3 = H_2 \left[M_{n \times n}^0 + M_{n \times n}^1 z + \dots + M_{n \times n}^k z^k \right] \quad (21)$$

where, H_3 specifies the output, $M_{n \times n}$ denotes the synaptic weight, and z assigns the element of z in the input vector, as well as H_2 denotes the output of the previous layer.

Now, substitute equation (20) in (21),

$$H_3 = A \left[\sum_{i=1}^n F_i + b_{\text{bias}} + \frac{1}{2} A \left(S \left(\sum_{j=1}^n \left(\sum_{k=1}^n \left(\sum_{l=1}^n E_{i,j,k} \right) \right) \right) \right) \right] M_{n \times n}^0 + M_{n \times n}^1 z + \dots + M_{n \times n}^k z^k \quad (22)$$

3.6.4 Training of DNF_VGG16 using HJSOA

The training procedure of DNF_VGG16 is acquired through the HJSOA, which is generated through the combination of SOA [40], and the Jaya Algorithm [39]. SOA is considered a human-based metaheuristic algorithm, which is motivated by individual efforts to develop their skills. Daily, the users of SOA attempt to enhance their skills through learning, but sometimes they fail to attain the optimal solution so the Jaya Algorithm [39] is utilized in this stage. The searching phase of the Jaya algorithm [39] usually moves away from the worst solution to achieve the best solution, which is utilized for several engineering crises. The modelling of the HJSOA is described below,

a) Initialization: The initial stage involves identifying the user within the search space, ensuring that the user is engaged in a random manner, which can be expressed as,

$$S = \begin{bmatrix} S_1 \\ \vdots \\ S_i \\ \vdots \\ S_N \end{bmatrix} = \begin{bmatrix} S_{1,1} & S_{1,d} & S_{1,m} \\ \vdots & \vdots & \vdots \\ S_{i,1} & S_{i,d} & S_{i,m} \\ \vdots & \vdots & \vdots \\ S_{N,1} & S_{N,d} & S_{N,m} \end{bmatrix} \quad (23)$$

where, S_i denotes the i^{th} user, $s_{i,d}$ indicates the d^{th} variable value through the i^{th} user of the entire population, N specifies the number of users (Population size), m represents the dimensionality of the optimization problem.

b) Fitness function

The fitness is calculated based on Mean Squared Error (MSE) to find an accurate solution, where the smallest MSE value is considered the best solution and is represented as,

$$U_o = \frac{1}{N} \sum_{v=1}^N (H_v^* - H_3)^2 \quad (24)$$

where, U_o denotes the fitness function, H_v^* and H_3 represents the expected outcome and predicted outcome of DNF_VGG16, N specifies the entire number of samples.

c) Exploration Stage

The population user attempts to improve their skill through a skilled user who consists of a high-quality state depending on the fitness value. The skilled user is chosen from the users with improved fitness values than the measured user in a random manner. It directs the population user to various positions in the search space through skill learning. The new spot is established when its fitness value is enhanced, which is uttered as,

$$s_{i,c} = 1 - s_{i,c} + r \cdot K_{i,c} - J \cdot s_{i,c} \quad (25)$$

$$s_{i,c} = 1 - s_{i,c} + r \cdot K_{i,c} - r \cdot J \cdot s_{i,c} \quad (26)$$

$$s_{i,c} = 1 - s_{i,c} + r \cdot J \cdot r \cdot K_{i,c} \quad (27)$$

where $s_{i,c}(0)$ represents the updated location of the user i,c in the initial stage, r indicates the uniformly distributed random number, which ranges from $[0,1]$, $K_{i,c}$ denotes the dimension of the skilled user, and J specifies the random number, which consists of the value as one or two.

To improve the performance of the solution, the updated equation of the Jaya algorithm [39] is substituted in the SOA [40]. From the Jaya algorithm,

$$s_{i,c}(0) = s_{i,c}(0) + r_1[s_b - |s_{i,c}(0)|] - r_2[s_w - |s_{i,c}(0)|] \quad (28)$$

Assign $s_{i,c}(0)$ as positive, and then equation (28) is rewritten as,

$$s_{i,c}(0) = s_{i,c}(0) + r_1[s_b - s_{i,c}(0)] - r_2[s_w - s_{i,c}(0)] \quad (29)$$

$$s_{i,c}(0) = s_{i,c}(0) + r_1[s_b - s_{i,c}(0)] - r_2[s_w - s_{i,c}(0)] \quad (30)$$

$$s_{i,c}(0) = \frac{s_{i,c}(0) + r_1[s_b - s_{i,c}(0)] - r_2[s_w - s_{i,c}(0)]}{1 - r_1 - r_2} \quad (31)$$

In the standard SOA, the candidate solution is updated according to the skilled-user learning mechanism. To improve the convergence behavior and avoid local optima, the best-worst learning strategy of the Jaya algorithm is incorporated into the SOA update process. Therefore, the Jaya update term shown in Equation (31) is embedded into the SOA position update equation (27). As a result, the updated position is influenced both by the skilled-user guidance of SOA and the best and worst solutions identified by Jaya. By substituting Equation (31) into Equation (27), the final HJSOA update equation is obtained as,

$$s_{i,c}(1) = s_{i,c}(0) + r_1[s_b - s_{i,c}(0)] - r_2[s_w - s_{i,c}(0)] + r[J - r]K_{i,c} \quad (32)$$

$$s_{i,c}(1) = s_{i,c}(0) + r_1[s_b - s_{i,c}(0)] - r_2[s_w - s_{i,c}(0)] + r[J - r]K_{i,c} \quad (33)$$

$$s_{i,c}(1) = \frac{1}{r[J - r]K_{i,c}} [s_{i,c}(0) + r_1[s_b - s_{i,c}(0)] - r_2[s_w - s_{i,c}(0)] + r[J - r]K_{i,c}] \quad (34)$$

where, s_w and s_b represents the worst and best solution, r indicates the random number, as well as K_i specifies the selected skilled user.

d) Exploitation Stage

Each **user's** efforts to develop the skills attained in the above phase are based on their preparation. This is modelled as a local search to raise each member and search for better circumstances to enhance the value of its fitness function. If the value of the fitness function is enhanced, subsequently the newly evaluated position is adequate. Thus, the solution is modelled as,

$$s_{i,c} = \begin{cases} s_{i,c} + \frac{1 - 2r}{l_o - u_o} (s_{i,c} - u_o) & r < 0.5 \\ s_{i,c} - \frac{1 - 2r}{l_o - u_o} (s_{i,c} - l_o) & \text{otherwise} \end{cases} \quad (35)$$

where, i represents the iteration counter, l_o and u_o denotes the lower as well as upper bound of a variable o .

e) Re-evaluation: The initial iteration is completed after the update of all users based on the exploration and exploitation phases. After this, the algorithm moves to the next iteration, and the updating process is repeated using Equations (23) to (35). The fitness of each solution is then evaluated at every iteration to determine the optimal solution.

f) Termination: The above-mentioned phases occur frequently however they **achieve** the optimal solution. The pseudo-code for the developed HJSOA is presented in **Algorithm 1**.

Algorithm 1: Pseudo code of HJSOA

Sl. No	Pseudo code of HJSOA
1	Begin
2	Initialization $S = \begin{bmatrix} S_1 \\ \vdots \\ S_i \\ \vdots \\ S_N \end{bmatrix} = \begin{bmatrix} s_{1,1} & s_{1,d} & s_{1,m} \\ \vdots & \vdots & \vdots \\ s_{i,1} & s_{i,d} & s_{i,m} \\ \vdots & \vdots & \vdots \\ s_{N,1} & s_{N,d} & s_{N,m} \end{bmatrix}$
3	Fitness function $U_o = \frac{1}{N} \sum_{v=1}^N (H_v^* - H_3)$
4	Exploration stage: Update the solution based on equation (34) $s_{i,c} = \begin{cases} 1 \\ \frac{1}{r \cdot J \cdot r_1 \cdot r_2} \cdot [r_2 s_w(\cdot) + r_1 s_b(\cdot) + (1 - r \cdot J) \cdot r \cdot K_{i,c} + r_1 \cdot r_2] \end{cases}$
5	Exploitation stage: Update the solution based on equation (35) $s_{i,c} = \begin{cases} s_{i,c} + \frac{1 - 2r}{\cdot} \cdot s_{i,c} \cdot r \cdot 0.5 \\ s_{i,c} + \frac{l_o \cdot r \cdot t_o \cdot l_o}{\cdot}, otherwise \end{cases}$
6	Re-evaluate to attain the best solution
7	End

Therefore, the DNF_VGG16 is trained by HJSOA, as well as attained the classified output.

4. Result and Discussion

This part discusses the results of a newly introduced approach, along with its evaluation on two datasets using validation metrics.

4.1 Experimental setup

The performance of the DNF_VGG16 model for BT classification is implemented using Python (version 3.9.11). The experiments are conducted on a system running Windows 11 with

8 GB RAM, over 100 GB of storage, a 1.7 GHz CPU, and GPU support for faster computation.

Table 2 illustrates the Experimental parameters associated with the developed DNF_VGG16 model.

Table 2. Experimental Parameters of the DNF_VGG16 model

Parameters	Values
DFVGG-16	
Epoch	80
Batch size	32
optimizer	HJSOA
kernel size	(3,3)
dropout	0.3
activation	Softmax
loss	categorical cross entropy
HJSOA	
N (Population size)	50
m(Dimension)	100
Maximum Iteration	80
l_o (lower bound)	0
u_o (Upper bound)	1
r	random numbers between (0,1)
J	randomly selected between (1 or 2)
Range of the Parameters	
Epoch	10 to 80
Batch size	16 or 32
dropout rate	0.2 to 0.5

4.2 Dataset description

The datasets applied for BT classification are the BraTS 2018 [31] and Figshare [32] datasets.

The dataset is split into 90% training, 5% testing, and 5% validation for effective evaluation of the proposed model, and the datasets are described below.

a) BraTS 2018 dataset

BraTS 2018 [31] employs preoperative MRI images from several organizations, which mainly focus on the segmentation of BTs such as gliomas that are essentially heterogeneous in their look, outline, as well as histology. In addition, this BraTS 2018 focuses on the calculation of

the entire survival rate of the patient, through techniques as well as integrative testing of radiomic characters to identify the clinical implications of this segmentation task.

b) Figshare dataset

Figshare dataset [32] contains 3064 T1-weighted contrast-improved images, which are obtained from 233 patients with 3 forms of BT. The Figshare dataset includes 708 meningioma images, 1426 glioma slices, and 930 pituitary tumor samples. It is divided into four parts based on file size, with each part containing 766 slices.

4.3 Evaluation measures

The performance of the DNF_VGG16 framework for BT classification is assessed using multiple accuracy measures, including TPR and TNR.

(i) **Accuracy** (A_c): It measures the division of every case that is classified accurately; it is the ratio of the number of exact classifications to the entire number of right or mistaken classifications, which is expressed as,

$$A_c = \frac{TP + TN}{TP + TN + FP + FN} \quad (36)$$

where, TP and TN specifies the true positive as well as true negative, whereas FP and FN denotes the false positive as well as false negative.

(ii) **TPR** (T_{pr}): It represents the proportion of correctly identified true positives by the proposed method out of the total actual true positives, and is expressed as,

$$T_{pr} = \frac{TP}{TP + FN} \quad (37)$$

where, T_{pr} denotes the true positive rate.

(iii) **TNR** (T_{nr}): It is defined as the proportion of true negatives correctly identified by the model relative to the total number of true negatives, which is expressed as,

$$T_{nr} = \frac{TP}{TP + FN} \quad (38)$$

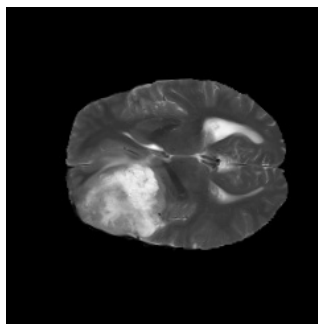
where, T_{nr} denotes the true negative rate.

4.4 Image results

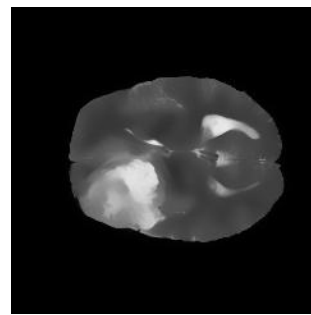
For BT classification, the image results of DNF_VGG16 with the images from both the BraTS 2018 [31] and Figshare [32] are discussed below.

4.4.1 Image results with BraTS 2018

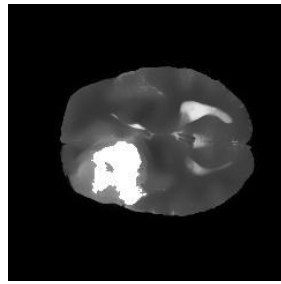
Figure 5 illustrates the results obtained from the BraTS 2018 dataset. Figure 5(a) represents the input image, while Figure 5(b) shows the pre-processed image. Figure 5(c) depicts the segmented image. The augmented outputs are presented as Figure 5(d) for the rotated image, Figure 5(e) for the scaled image, Figure 5(f) for the color-augmented image, Figure 5(g) for the contrast-enhanced image, and Figure 5(h) for the translated image. Figure 5(i) shows the LTxXORP features extracted from the rotated image, Figure 5(j) shows the LTxXORP features extracted from the scaled image, Figure 5(k) shows the LTxXORP features extracted from the color-augmented image, Figure 5(l) shows the LTxXORP features extracted from the contrast-enhanced image, and Figure 5(m) shows the LTxXORP features extracted from the translated image.



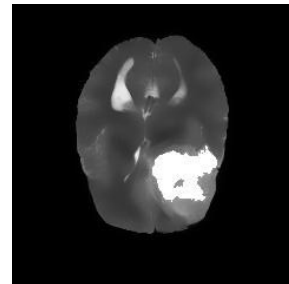
(a)



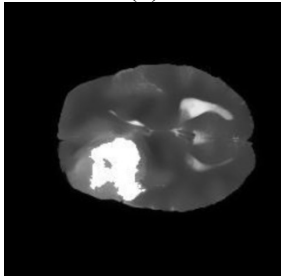
(b)



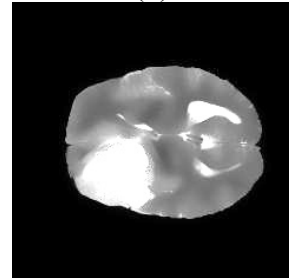
(c)



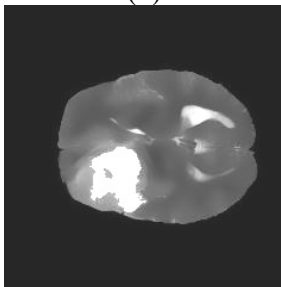
(d)



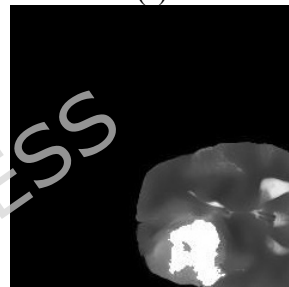
(e)



(f)



(g)



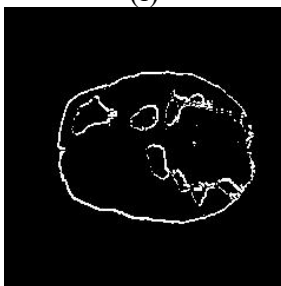
(h)



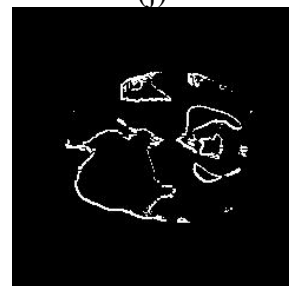
(i)



(j)



(k)



(l)

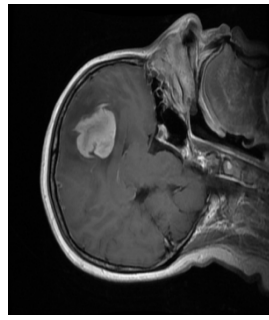


(m)

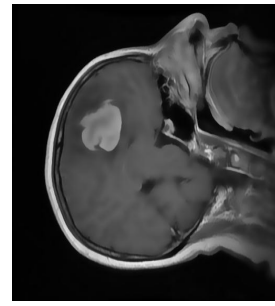
Figure 5. Image results with BraTS 2018: a) input image, b) pre-processed image, c) segmented image, d) rotated image, e) scaled image, f) color augmented image, g) contrasted image, h) translated image, i) rotated LTxXORP features, j) scaled LTxXORP features, k) color augmented LTxXORP features, l) contrasted LTxXORP features, and m) translated LTxXORP features

4.4.2 Image results with Figshare

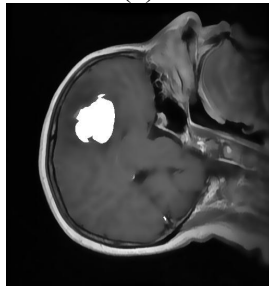
Figure 6 illustrates the results obtained from the Figshare dataset. Figure 6(a) represents the input image, while Figure 6(b) shows the pre-processed image. Figure 6(c) depicts the segmented image. The augmented outputs are presented as Figure 6(d) for the rotated image, Figure 6(e) for the scaled image, Figure 6(f) for the color-augmented image, Figure 6(g) for the contrast-enhanced image, and Figure 6(h) for the translated image. Furthermore, Figure 6(i) shows the LTxXORP features extracted from the rotated image, Figure 6(j) shows the LTxXORP features extracted from the scaled image, Figure 6(k) shows the LTxXORP features extracted from the color-augmented image, Figure 6(l) shows the LTxXORP features extracted from the contrast-enhanced image, and Figure 6(m) shows the LTxXORP features extracted from the translated image.



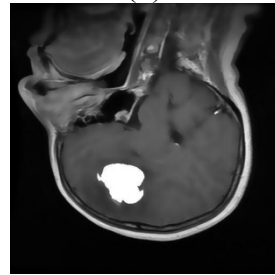
(a)



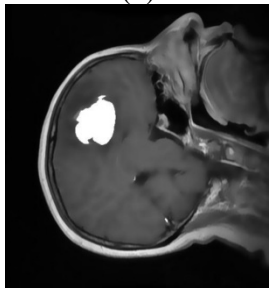
(b)



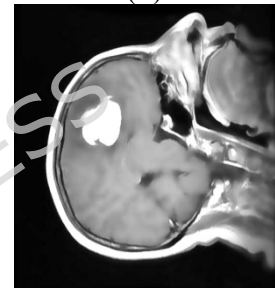
(c)



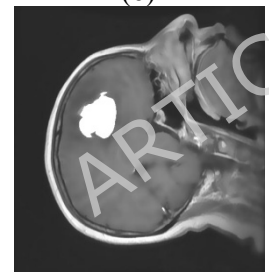
(d)



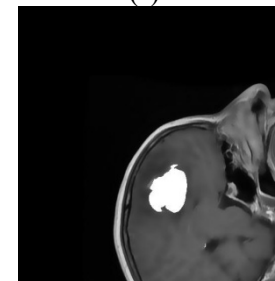
(e)



(f)



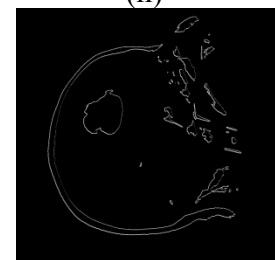
(g)



(h)



(i)



(j)

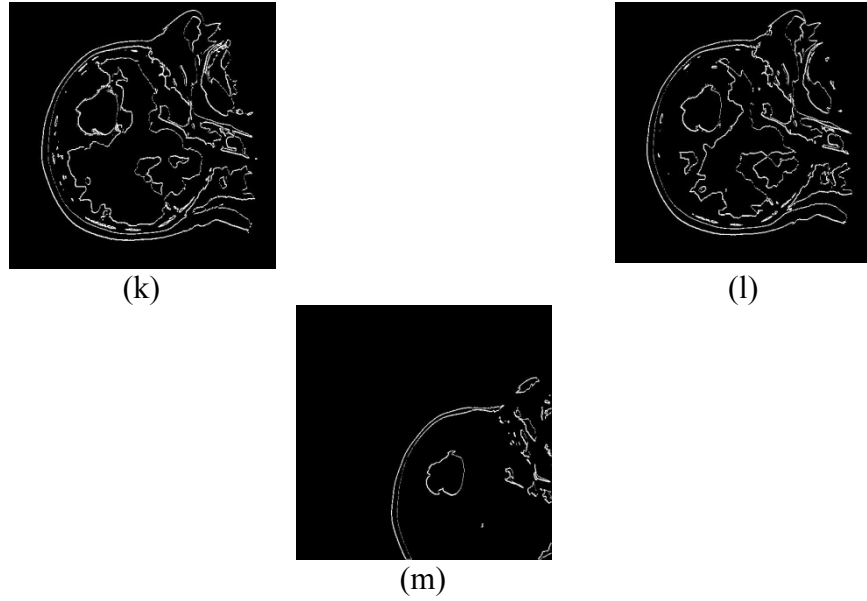


Figure 6. Image results with Figshare: a) input image, b) pre-processed image, c) segmented image, d) rotated image, e) scaled image, f) color augmented image, g) contrasted image, h) translated image, i) rotated LTxXORP features, j) scaled LTxXORP features, k) color augmented LTxXORP features, l) contrasted LTxXORP features, and m) translated LTxXORP features

4.5 Algorithm Analysis

The analysis of algorithms is primarily employed to assess the performance of the model in terms of its time and space complexity. The algorithm analysis is done with two datasets such as BraTS 2018 and Figshare for the developed HJSOA+DNF_VGG16 over the existing methods of Genetic Algorithm (GA) [46]+DNF_VGG16, Tabu Search (TS) [47]+DNF_VGG16, Adam [48]+DNF_VGG16, Jaya [39]+DNF_VGG16, Particle Swarm Optimization (PSO) [49]+DNF_VGG16, SOA [40]+DNF_VGG16, Ant colony optimization (ACO) [50] +DNF_VGG16, and Ensemble (Jaya+SOA)+DNF_VGG16. In the ensemble approach, Jaya and SOA were executed independently, and the final performance was obtained by averaging their results. This comparison was included to investigate whether the

performance improvement achieved by HJSOA originates from the proposed hybridization mechanism or merely from combining two optimization algorithms.

4.5.1 With BraTS 2018

Figure 7 shows the algorithm analysis of the developed method of HJSOA+DNF_VGG16 with BraTS 2018 over the existing approaches, such as GA+DNF_VGG16, TS+DNF_VGG16, Adam+DNF_VGG16, Jaya+DNF_VGG16, PSO+DNF_VGG16, SOA+DNF_VGG16, ACO+DNF_VGG16, and Ensemble (Jaya+SOA)+DNF_VGG16, done by varying the iterations 20, 40, 60, and 80. At iteration 80, the accuracy of the developed HJSOA+DNF_VGG16 model is 0.912, which is greater than the other previous methods such as GA+DNF_VGG16 is 0.835, TS+DNF_VGG16 is 0.858, Adam+DNF_VGG16 is 0.867, Jaya+DNF_VGG16 is 0.872, PSO+DNF_VGG16 is 0.885, SOA+DNF_VGG16 is 0.890, ACO+DNF_VGG16 is 0.903, and Ensemble (Jaya+SOA)+DNF_VGG16 is 0.907 as revealed in Figure 7 a). Similarly, the algorithm analysis of the TPR for the HJSOA+DNF_VGG16 model is represented in Figure 7 b). Here, the value of TPR for the system of HJSOA+DNF_VGG16 at iteration 80 is 0.936, which is better than the previous approaches like GA+DNF_VGG16, TS+DNF_VGG16, Adam+DNF_VGG16, Jaya+DNF_VGG16, PSO+DNF_VGG16, SOA+DNF_VGG16, ACO+DNF_VGG16, and Ensemble (Jaya+SOA)+DNF_VGG16 is 0.856, 0.873, 0.886, 0.896, 0.905, 0.911, 0.923, and 0.928, respectively. Finally, the TNR value for the HJSOA+DNF_VGG16 system at iteration 80 is 0.924, which is superior to the GA+DNF_VGG16 is 0.840, TS+DNF_VGG16 is 0.860, Adam+DNF_VGG16 is 0.874, Jaya+DNF_VGG16 is 0.885, PSO+DNF_VGG16 is 0.891, SOA+DNF_VGG16 is 0.896, ACO+DNF_VGG16 is 0.904, and Ensemble (Jaya+SOA)+DNF_VGG16 is 0.916, which is represented in figure 7 c).

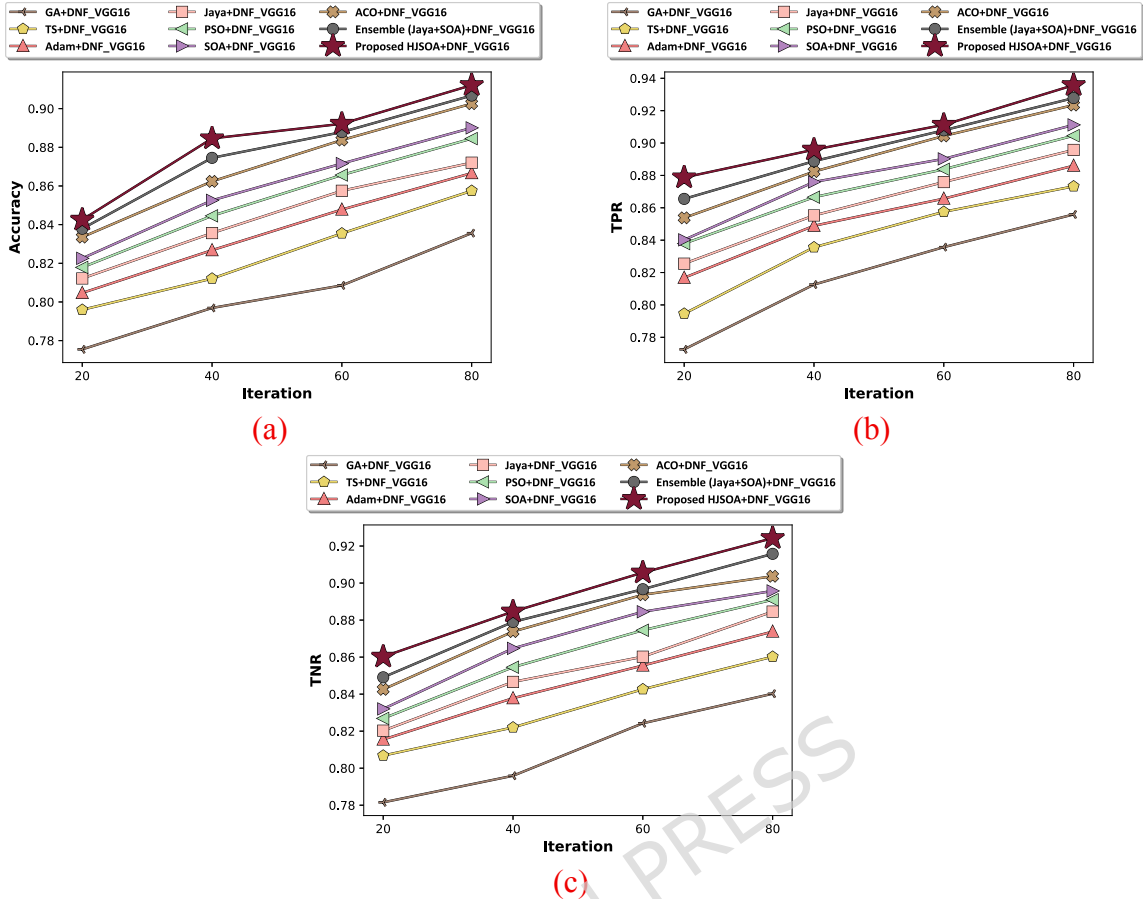


Figure 7. Algorithm analysis of HJSOA+DNF_VGG16 with BraTS 2018: a) Accuracy, b) TPR, c) TNR

4.5.2 With Figshare

Figure 8 presents the algorithmic analysis of the proposed HJSOA+DNF_VGG16 model on the Figshare dataset, evaluated by varying the number of iterations and comparing it with existing approaches, such as GA+DNF_VGG16, TS+DNF_VGG16, Adam+DNF_VGG16, Jaya+DNF_VGG16, PSO+DNF_VGG16, SOA+DNF_VGG16, ACO+DNF_VGG16, and Ensemble (Jaya+SOA)+DNF_VGG16. Figure 8 a) illustrates the accuracy level of the developed model by altering the iterations. The accuracy of HJSOA+DNF_VGG16 obtained the value of 0.912 at iteration 80, whereas the existing method of GA+DNF_VGG16 acquired 0.830, TS+DNF_VGG16 attained 0.858, Adam+DNF_VGG16 obtained 0.863, Jaya+DNF_VGG16 acquired 0.876, PSO+DNF_VGG16 attained 0.882, SOA+DNF_VGG16

acquired 0.892, ACO+DNF_VGG16 obtained 0.904, and Ensemble (Jaya+SOA)+DNF_VGG16 obtained 0.908. At iteration 60, the value of TPR for the developed approach of HJSOA+DNF_VGG16 is 0.895%, which is better than the GA+DNF_VGG16, TS+DNF_VGG16, Adam+DNF_VGG16, Jaya+DNF_VGG16, PSO+DNF_VGG16, SOA+DNF_VGG16, ACO+DNF_VGG16, and Ensemble (Jaya+SOA)+DNF_VGG16 is 0.812, 0.835, 0.843, 0.856, 0.865, 0.873, 0.883, and 0.889, respectively, as illustrated in Figure 8 b). Lastly, Figure 8 c) denotes the value of TNR with Figshare. Here, the HJSOA+DNF_VGG16 achieves 0.929 at iteration 80, whereas GA+DNF_VGG16 is 0.835, TS+DNF_VGG16 is 0.856, Adam+DNF_VGG16 is 0.863, Jaya+DNF_VGG16 is 0.875, PSO+DNF_VGG16 is 0.885, SOA+DNF_VGG16 is 0.890, ACO+DNF_VGG16 is 0.903, and Ensemble (Jaya+SOA)+DNF_VGG16 is 0.915.

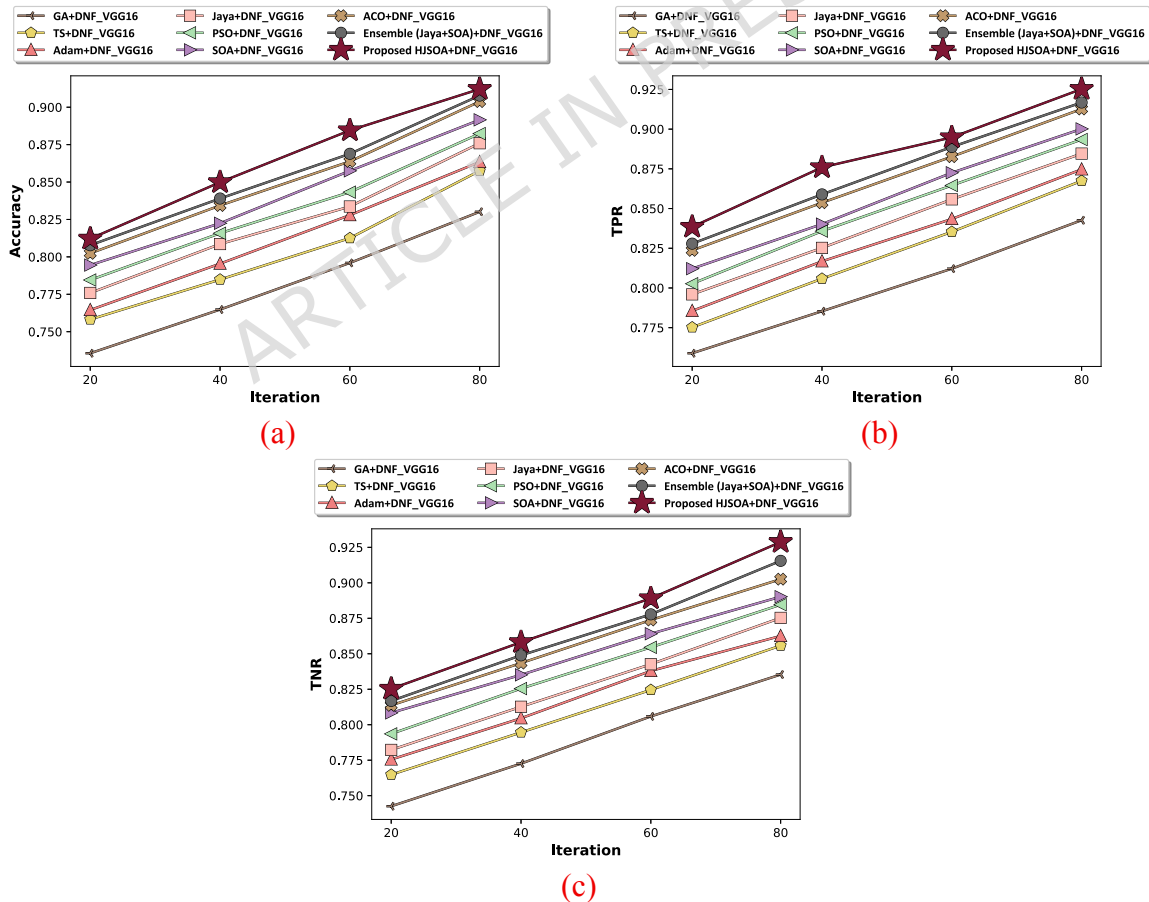


Figure 8. Algorithm analysis of HJSOA+DNF_VGG16 with Figshare a) Accuracy, b) TPR, c) TNR

4.6 Comparative methods

The developed HJSOA+DNF_VGG16 scheme is compared with other preceding systems such as DCNN [51], MAG-Net [52], Hybrid CNN-SVM [53], DNN [54], CAHO-based DNFN [55], ResNet101+GWO+Multi-Transformer Fusion model [56], DNF_VGG16, ConvNeXt Base model [57], and TCAFNet [58].

4.7 Comparative analysis

The comparative analysis is conducted by modifying the learning sets for the designated images.

4.7.1 Estimation with BraTS 2018

Table 3 presents the analysis of the proposed HJSOA+DNF_VGG16 model on the BraTS 2018 dataset by varying the learning sets. The HJSOA+DNF_VGG16 method achieves an accuracy of 0.930 at a 90% learning set. In comparison, other methods such as DCNN, MAG-Net, Hybrid CNN-SVM, DNN, CAHO-based DNFN, ResNet101+GWO+Multi-Transformer Fusion model, DNF_VGG16, ConvNeXt Base model, and TCAFNet obtain accuracy of 0.772, 0.796, 0.823, 0.858, 0.885, 0.898, 0.916, 0.923, and 0.927, respectively. The TPR of the proposed HJSOA+DNF_VGG16 model at a 90% learning set is 0.950. In comparison, the TPR values of existing methods such as DCNN, MAG-Net, Hybrid CNN-SVM, DNN, CAHO-based DNFN, ResNet101+GWO+Multi-Transformer Fusion model, DNF_VGG16, ConvNeXt Base model, and TCAFNet are 0.795, 0.813, 0.842, 0.885, 0.906, 0.925, 0.930, 0.935, and 0.939, respectively. The TNR value of the proposed HJSOA+DNF_VGG16 model is 0.940, which outperforms existing methods. In comparison, the TNR values of conventional models are 0.783, 0.809, 0.830, 0.862, 0.885, 0.917, 0.923, 0.925, and 0.928, respectively.

Table 3. Comparative analysis with BraTS 2018 based on learning sets using Accuracy, TPR, and TNR.

Metri cs	Lear ning set	DCN N	MAG- Net	Hybrid CNN- SVM	DNN	CAHO- based DNFN	ResNet101 + GWO + Multi- Transformer Fusion model	DNF VG G16	ConvN eXt Base model	TCA FNet	Proposed HJSOA+ DNF_VG G16
Accu racy	60	0.693	0.711	0.743	0.782	0.811	0.828	0.843	0.852	0.859	0.864
	70	0.711	0.731	0.761	0.796	0.842	0.857	0.875	0.884	0.888	0.897
	80	0.759	0.773	0.809	0.836	0.860	0.884	0.895	0.903	0.907	0.914
	90	0.772	0.796	0.823	0.858	0.885	0.898	0.916	0.923	0.927	0.930
TPR	60	0.712	0.731	0.763	0.806	0.830	0.849	0.867	0.874	0.879	0.880
	70	0.745	0.786	0.811	0.840	0.873	0.887	0.898	0.900	0.902	0.903
	80	0.762	0.809	0.843	0.862	0.885	0.898	0.912	0.925	0.928	0.931
	90	0.795	0.813	0.842	0.885	0.906	0.925	0.930	0.935	0.939	0.950
TNR	60	0.695	0.713	0.759	0.795	0.825	0.843	0.855	0.864	0.869	0.873
	70	0.759	0.779	0.809	0.836	0.863	0.875	0.885	0.894	0.898	0.903
	80	0.776	0.796	0.820	0.842	0.874	0.895	0.907	0.910	0.914	0.920
	90	0.783	0.809	0.830	0.862	0.885	0.917	0.923	0.925	0.928	0.940

4.7.2 Estimation with Figshare

The analysis of HJSOA+DNF_VGG16 based on Figshare is estimated by varying learning sets are shown in Table 4. At the learning set of 90, the accuracy value of the developed HJSOA+DNF_VGG16 is **0.928**, which is 14.30% better than DCNN, 12.48% better than MAG-Net, 10.44% better than Hybrid CNN-SVM, 6.56% better than DNN%, 4.58% better than CAHO-based DNFN, **2.67% better than ResNet101+GWO+Multi-Transformer Fusion model**, 1.82% better than DNF_VGG16, **1.47% better than ConvNeXt Base model**, and **1.15% better than TCAFNNet**. The TPR of the developed HJSOA+DNF_VGG16 acquired the value of **0.944** at the learning set 90, which is improved than the previous methods of DCNN with a value of 12.55%, MAG-Net with 10.71%, Hybrid CNN-SVM with 8.62%, DNN with 6.27%, CAHO-based DNFN with 4.47%, **ResNet101+GWO+Multi-Transformer Fusion model with 3.08%**, DNF_VGG16 with 2.49%, **ConvNeXt Base model with 0.95%**, and TCAFNNet with **0.63%**. The value of TNR for the HJSOA+DNF_VGG16 acquired **0.936**, which is 15.14% better than DCNN, 12.41% better than MAG-Net, 10.08% better than Hybrid CNN-SVM, 7.87% better than DNN%, 5.44% better than CAHO-based DNFN, **4.24% better than**

ResNet101+GWO+Multi-Transformer Fusion model, 2.57% better than DNF_VGG16, 1.24% better than ConvNeXt Base model, and 1.02% better than TCAFNet.

Table 4. Comparative analysis with Figshare based on learning sets using Accuracy, TPR, and TNR.

Metric s	Lear ning set	DCN N	MAG- Net	Hybrid CNN- SVM	DNN	CAHO- based DNFN	ResNet101 + GWO + Multi- Transformer Fusion model	DNF_ VGG1 6	ConvN eXt Base model	TCA FNet	Proposed HJSOA+ DNF_VG G16
Accura cy	60	0.709	0.725	0.742	0.779	0.796	0.806	0.812	0.824	0.827	0.835
	70	0.712	0.735	0.768	0.800	0.813	0.833	0.842	0.854	0.858	0.867
	80	0.790	0.797	0.820	0.843	0.856	0.863	0.876	0.879	0.879	0.880
	90	0.796	0.813	0.832	0.867	0.886	0.904	0.912	0.915	0.918	0.928
TPR	60	0.720	0.743	0.756	0.763	0.809	0.825	0.832	0.842	0.847	0.857
	70	0.754	0.779	0.800	0.820	0.843	0.854	0.867	0.874	0.878	0.884
	80	0.785	0.809	0.836	0.858	0.870	0.886	0.895	0.904	0.908	0.914
	90	0.825	0.843	0.862	0.885	0.902	0.915	0.920	0.935	0.938	0.944
TNR	60	0.712	0.735	0.740	0.759	0.795	0.809	0.825	0.835	0.838	0.841
	70	0.743	0.760	0.786	0.806	0.820	0.843	0.853	0.863	0.868	0.873
	80	0.770	0.795	0.812	0.831	0.862	0.874	0.885	0.893	0.898	0.903
	90	0.795	0.820	0.842	0.863	0.885	0.897	0.912	0.925	0.927	0.936

4.8 Statistical Analysis

Statistical analysis helps determine whether the performance improvements achieved by a proposed model are meaningful or arise due to random variations in the experimental process. In this work, statistical analysis was performed using Mean $\pm$ Standard Deviation (SD) and a paired t-test.

4.8.1 Paired T-test Analysis

The paired t-test is a statistical method used to determine whether the difference between two related sets of observations is statistically significant. In this work, the paired t-test was employed to evaluate whether the performance improvements achieved by the proposed HJSOA+DNF_VGG16 model over existing methods are statistically significant or merely due to random variations. The analysis was performed using the Accuracy, TPR, and TNR values

obtained from multiple experimental runs on the BraTS 2018 and Figshare datasets. A significance level of 0.05 was adopted, where p-values less than 0.05 indicate statistically significant differences. The paired t-test results presented in Table 5 indicate that the proposed HJSOA+DNF_VGG16 model consistently outperforms all comparative methods across both datasets. For the BraTS 2018 dataset, statistically significant improvements in classification accuracy were observed over DCNN ($t = 2.889$, $p = 0.017$), MAG-Net ($t = 2.568$, $p = 0.020$), Hybrid CNN-SVM ($t = 2.118$, $p = 0.026$), DNN ($t = 1.988$, $p = 0.030$), CAHO-based DNFN ($t = 1.799$, $p = 0.033$), ResNet101 + GWO + Multi-Transformer Fusion model ($t = 1.698$, $p = 0.035$), DNF_VGG16 ($t = 1.590$, $p = 0.038$), ConvNeXt Base model ($t = 1.510$, $p = 0.040$), and TCAFNet ($t = 1.488$, $p = 0.043$). Furthermore, the TPR and TNR comparisons also yielded p-values below 0.05, confirming statistically significant improvements in sensitivity and specificity. For Figshare dataset, the proposed HJSOA+DNF_VGG16 model exhibited significant performance gains over DCNN ($t = 3.104$, $p = 0.014$), MAG-Net ($t = 2.783$, $p = 0.017$), Hybrid CNN-SVM ($t = 2.333$, $p = 0.023$), DNN ($t = 2.204$, $p = 0.027$), CAHO-based DNFN ($t = 2.015$, $p = 0.029$), ResNet101 + GWO + Multi-Transformer Fusion model ($t = 1.914$, $p = 0.032$), DNF_VGG16 ($t = 1.806$, $p = 0.035$), ConvNeXt Base model ($t = 1.725$, $p = 0.037$), and TCAFNet ($t = 1.703$, $p = 0.040$). The corresponding TPR and TNR results also produced p-values lower than the significance threshold, indicating statistically significant improvements across all evaluation metrics. Since all obtained p-values are less than 0.05, the null hypothesis is rejected for all comparisons. Therefore, the observed improvements achieved by the proposed HJSOA+DNF_VGG16 model are statistically significant and demonstrate the effectiveness and robustness of the proposed framework for BT classification.

Table 5. Paired T-Test Results of the proposed HJSOA+DNF_VGG16

Proposed Method	Comparative Methods	BraTS 2018 dataset					
		T-statistics			P-value		
		Accuracy	TPR	TNR	Accuracy	TPR	TNR
Proposed HJSOA+DNF_VGG16	DCNN	2.889	3.249	2.589	0.017	0.015	0.020
	MAG-Net	2.568	2.989	2.379	0.020	0.019	0.024
	Hybrid CNN-SVM	2.118	2.590	1.988	0.026	0.023	0.029
	DNN	1.988	2.379	1.799	0.030	0.026	0.033
	CAHO-based DNFN	1.799	2.099	1.668	0.033	0.030	0.036
	ResNet101 + GWO + Multi-Transformer Fusion model	1.698	1.989	1.479	0.035	0.033	0.040
	DNF_VGG16	1.590	1.888	1.380	0.038	0.036	0.043
	ConvNeXt Base model	1.510	1.689	1.319	0.040	0.038	0.046
	TCAFFNet	1.488	1.579	1.288	0.043	0.040	0.047
Figshare dataset							
Proposed HJSOA+DNF_VGG16	DCNN	3.104	3.465	2.805	0.014	0.012	0.017
	MAG-Net	2.783	3.204	2.595	0.017	0.016	0.021
	Hybrid CNN-SVM	2.333	2.806	2.204	0.023	0.019	0.026
	DNN	2.204	2.595	2.015	0.027	0.023	0.030
	CAHO-based DNFN	2.015	2.315	1.883	0.029	0.027	0.033
	ResNet101 + GWO + Multi-Transformer Fusion model	1.914	2.204	1.695	0.032	0.030	0.037
	DNF_VGG16	1.806	2.104	1.596	0.035	0.033	0.040
	ConvNeXt Base model	1.725	1.905	1.535	0.037	0.035	0.043
	TCAFFNet	1.703	1.795	1.503	0.040	0.037	0.044

4.8.2 Mean±SD

Mean $\pm$ SD analysis is a commonly used statistical method in deep learning to evaluate both the overall performance and stability of a model across multiple independent runs. The mean represents the average value of a performance metric, while the SD indicates the variation or consistency of the results. Table 6 shows the Mean $\pm$ SD analysis of the proposed model. A higher mean with a lower standard deviation indicates that the model performs well with stable and consistent outcomes. For the BraTS 2018 dataset, the proposed HJSOA+DNF_VGG16 model achieved mean values of 0.901 for Accuracy, 0.916 for TPR, and 0.909 for TNR. The corresponding standard deviation values were 0.028, 0.025, and 0.021, respectively. These results show that the proposed model provides high classification performance with low variability across multiple runs, indicating strong stability and reliability. Similarly, for the

Figshare dataset, the proposed model obtained mean values of 0.877 for Accuracy, 0.899 for TPR, and 0.888 for TNR, with standard deviation values of 0.038, 0.037, and 0.034, respectively. These results further confirm that the model produces consistent performance across different experimental runs. The results demonstrate that the proposed HJSOA+DNF_VGG16 model achieves both high accuracy and stable performance on both datasets.

Table 6. Mean $\pm$ SD Results of the HJSOA+DNF_VGG16 model

Comparative Methods /Metrics	BraTS 2018 dataset						Figshare dataset					
	Mean			SD			Mean			SD		
	Accur acy	TP R	TN R	Accur acy	TP R	TN R	Accur acy	TP R	TN R	Accur acy	TP R	TN R
DCNN	0.733	0.753	0.752	0.038	0.035	0.040	0.750	0.770	0.754	0.048	0.045	0.036
MAG-Net	0.752	0.784	0.773	0.039	0.037	0.043	0.767	0.792	0.777	0.044	0.043	0.037
Hybrid CNN-SVM	0.783	0.814	0.804	0.038	0.038	0.032	0.790	0.812	0.794	0.042	0.046	0.043
DNN	0.817	0.848	0.833	0.035	0.034	0.028	0.821	0.830	0.814	0.040	0.053	0.044
CAHO-based DNFN	0.849	0.873	0.861	0.031	0.032	0.026	0.837	0.855	0.840	0.041	0.040	0.041
ResNet101 + GWO + Multi-Transformer Fusion model	0.866	0.889	0.882	0.031	0.032	0.031	0.851	0.869	0.855	0.042	0.039	0.038
DNF_VGG16	0.882	0.902	0.892	0.031	0.026	0.029	0.860	0.878	0.868	0.043	0.038	0.038
ConvNeXt Base model	0.890	0.908	0.898	0.030	0.027	0.026	0.867	0.888	0.878	0.039	0.040	0.039
TCAFFNet	0.895	0.911	0.902	0.029	0.027	0.025	0.870	0.892	0.882	0.038	0.039	0.038
proposed HJSOA+DNF_VGG16	0.901	0.916	0.909	0.028	0.025	0.021	0.877	0.899	0.888	0.038	0.037	0.034

4.9 Ablation Study Analysis

An ablation study is an experimental evaluation technique used to measure the contribution of each component of a proposed model by systematically removing or modifying one component at a time while keeping the remaining framework unchanged. This helps to identify the importance and effectiveness of each module in the overall system performance. Figure 9

presents the ablation study analysis of the proposed HJSOA–DNF_VGG16 model for both the BraTS 2018 and Figshare datasets. The study evaluates the contribution of different components, including WFKM segmentation, image augmentation, feature extraction, bilateral filtering, and different model configurations such as DNFN, VGG16, DNF_VGG16, Jaya-DNF_VGG16, and SOA-DNF_VGG16. Figure 9 a) shows the ablation analysis concerning BraTS 2018 dataset. When the learning set is considered as 90%, the proposed HJSOA+DNF_VGG16 model achieved an accuracy of 0.930, which is higher than all other ablated configurations. The accuracy values for different configurations are 0.835 for the model without WFKM, 0.859 for without image augmentation, 0.867 for without feature extraction, 0.880 for without bilateral filtering, 0.893 for DNFN, 0.908 for VGG16, 0.916 for DNF_VGG16, 0.924 for Jaya-DNF_VGG16, and 0.928 for SOA-DNF_VGG16. Figure 9 b) shows the ablation results concerning Figshare dataset. At a 90% learning set, the proposed model achieved an accuracy of 0.928, while the ablated configurations obtained lower performance values of 0.828 for without WFKM, 0.838 for without image augmentation, 0.849 for without feature extraction, 0.859 for without bilateral filtering, 0.878 for DNFN, 0.888 for VGG16, 0.895 for DNF_VGG16, 0.907 for Jaya-DNF_VGG16, and 0.918 for SOA-DNF_VGG16. These results further confirm that each module plays an important role in improving the final classification performance.

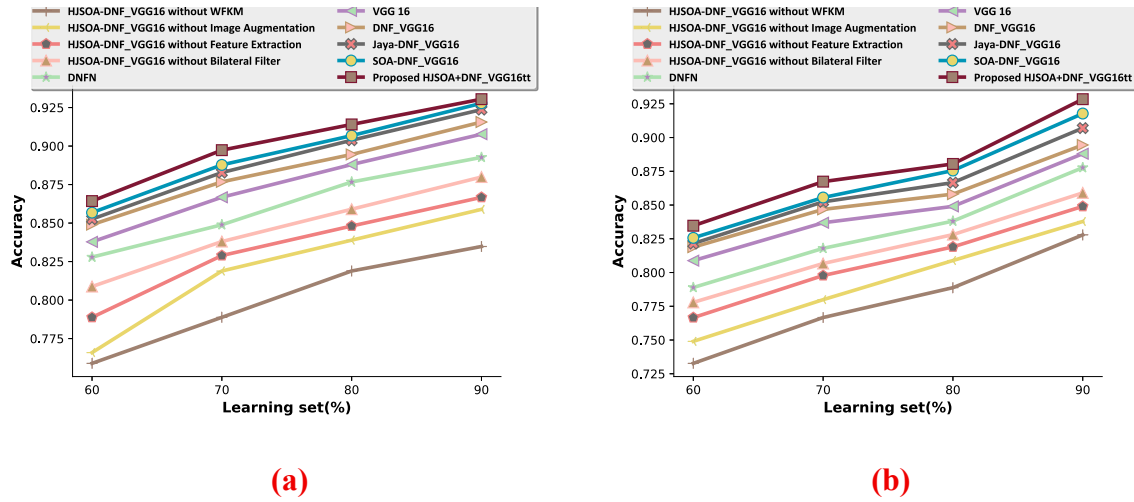


Figure 9. Ablation Analysis of the Proposed HJSOA+DNF_VGG16 model, a) BraTS 2018 dataset and b) Figshare dataset

4.10 Limitations

The major limitations of the devised model are,

- The proposed HJSOA_DNF_VGG16 model cannot generalize well across other medical image modalities, such as CT scans or endoscopic images.
- This method may struggle to identify tumors with irregular spatial distributions.
- This method cannot adapt to dynamic data evolution caused by changes in imaging protocols, disease characteristics, and patient demographics.
- The employed BraTS 2018 and Figshare datasets do not contain tumor-stage annotations or longitudinal disease progression information. Therefore, the proposed HJSOA_DNF_VGG16 model was evaluated only for BT classification performance and not for stage-specific or early-stage tumor detection.

5. Conclusion

In this research, the HJSOA+based DNF_VGG16 is developed to classify the BT with the MRI images. Here, the MRI image serves as the input, with noise removal performed through a Bilateral filter, followed by segmentation using WFKM. The segmented images are then

augmented with the techniques of rotation, scaling, color augmentation, contrast, and translation. Subsequently, these augmented images are then extracted through DWT with LTxXORP. The extracted features are further classified using the DNF_VGG16 and trained using the HJSOA. Therefore, the performance of the HJSOA+DNF_VGG16 scheme is analyzed by varying the learning sets via the BRATS 2018 as well as the Figshare datasets. In the dataset of BRATS 2018, the developed HJSOA+DNF_VGG16 model attained the values of 0.930 for accuracy, 0.950 for TPR, and 0.940 for TNR at the learning sets of 90. Similarly, at the learning sets of 90, the values of accuracy, TPR, and TNR of the developed model for the Figshare dataset are 0.928, 0.944, and 0.936. In the future, multiple images should be considered to create better outcomes. The devised model will be applied to more recent and diverse datasets to validate the generalization of the model. Additionally, the proposed method will be expanded to generalize well on other medical image modalities such as CT scans and endoscopic images.

Declaration Statements:

Funding: This research did not receive any specific funding

Conflict of Interest: The authors declare no conflict of interest

Acknowledgements: I would like to express my very great appreciation to the co-authors of this manuscript for their valuable and constructive suggestions during the planning and development of this research work.

Informed consent: Not Applicable

Ethical approval: Not Applicable

Author Contribution: Naga Raju Arumalla designed the model and the computational framework and analysed the data. Veerraju Gampala carried out the implementation. All

authors performed the calculations. All authors wrote the manuscript with input from all authors. All authors conceived the study and were in charge of overall direction and planning.

Data Availability Statement:

The data underlying this article are available in *BRATS 2018 database* is collected from “<https://wiki.cancerimagingarchive.net/pages/viewpage.action?pageId=37224922>” and the *Figshare dataset* is collected from <https://www.kaggle.com/datasets/ashkhagan/figshare-brain-tumor-dataset>.

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